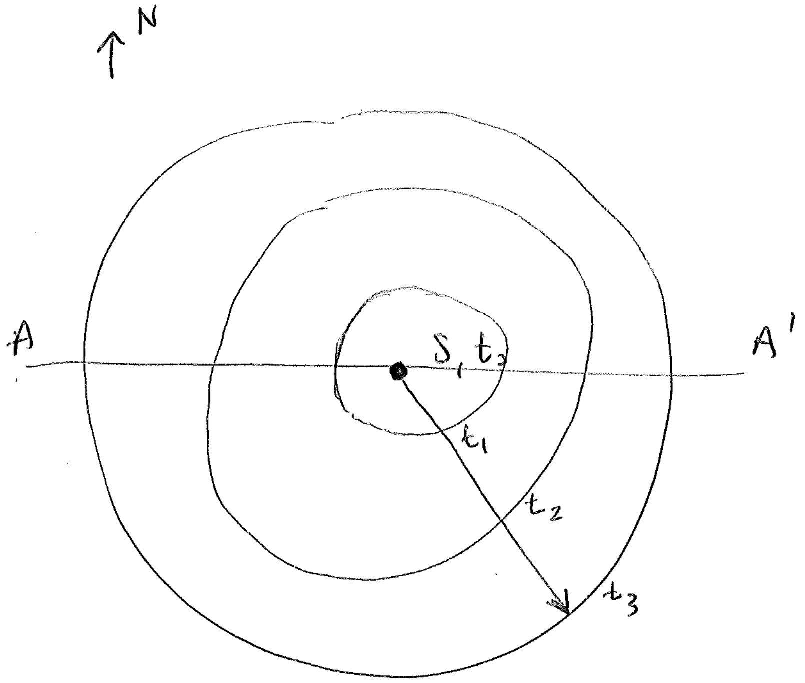
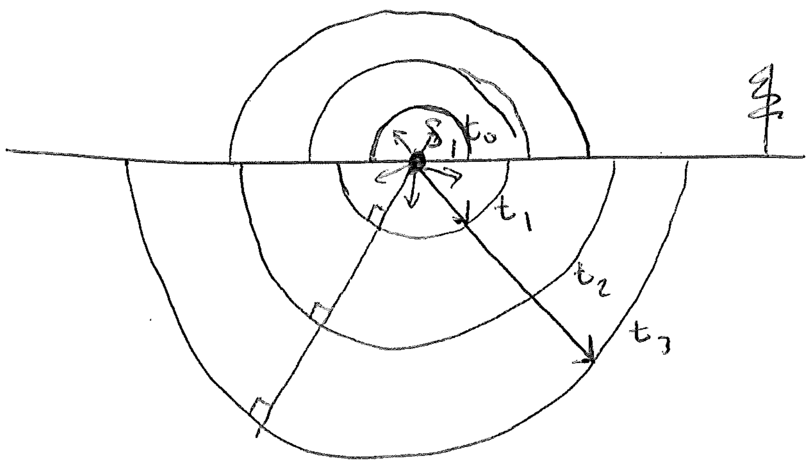
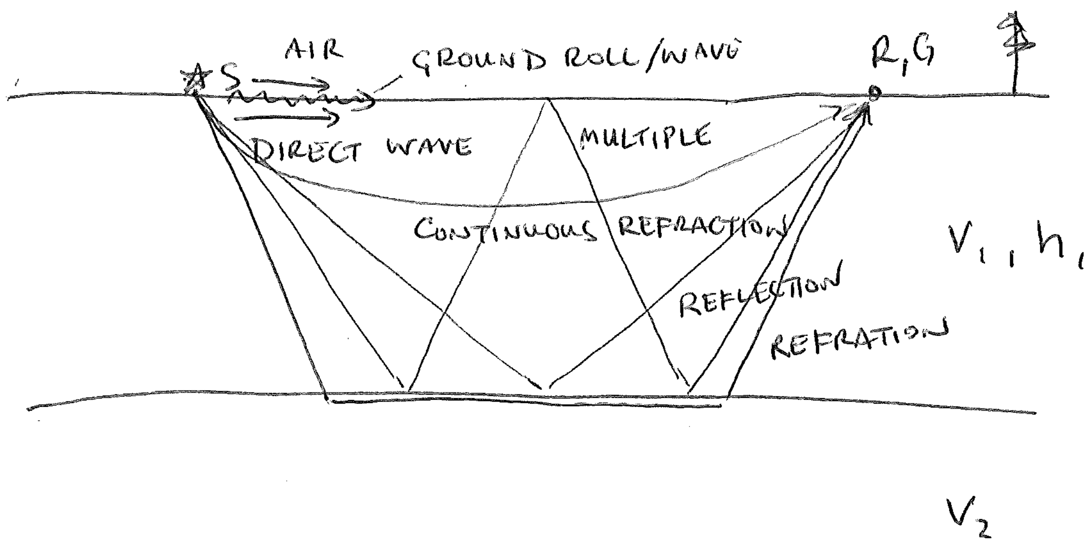
REFLECTION/REFRACTION SEISMOLOGY 2014/2015

MAP VIEW





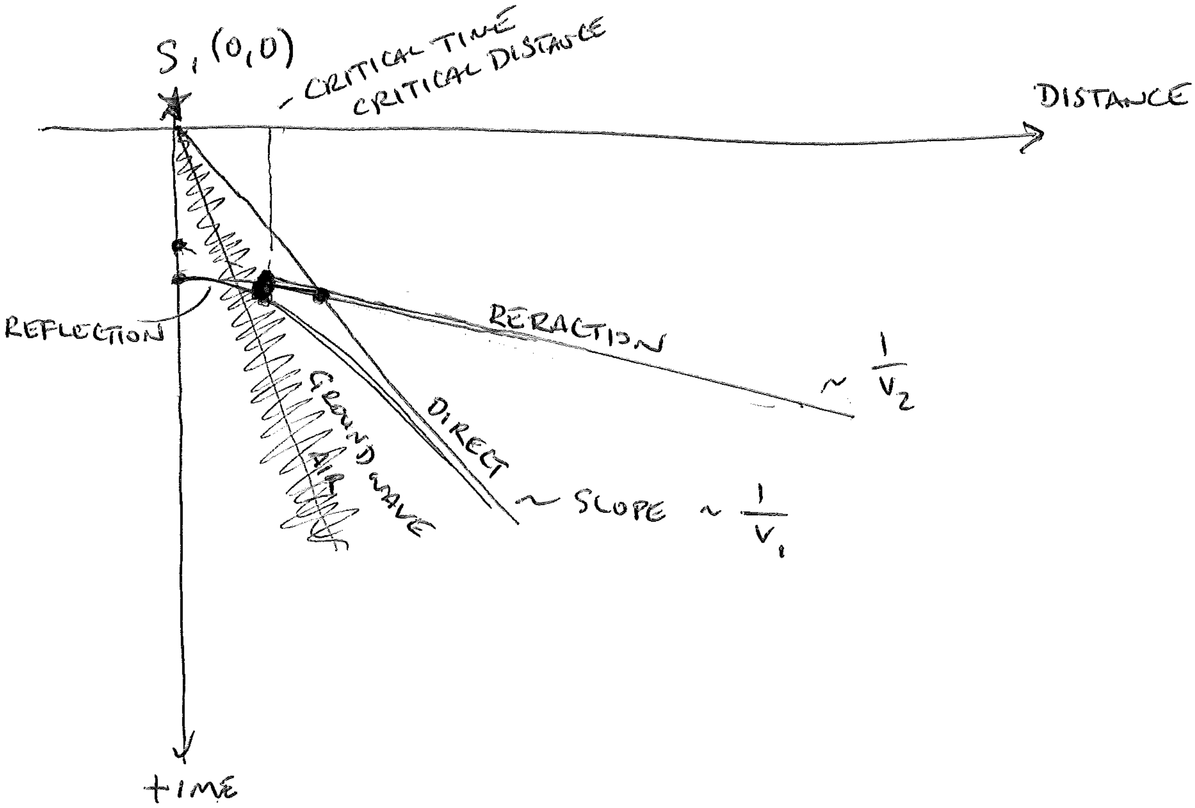
$$l_{WAVEFRONT} = t_{ELAPSED} V_{MEDIUM}$$

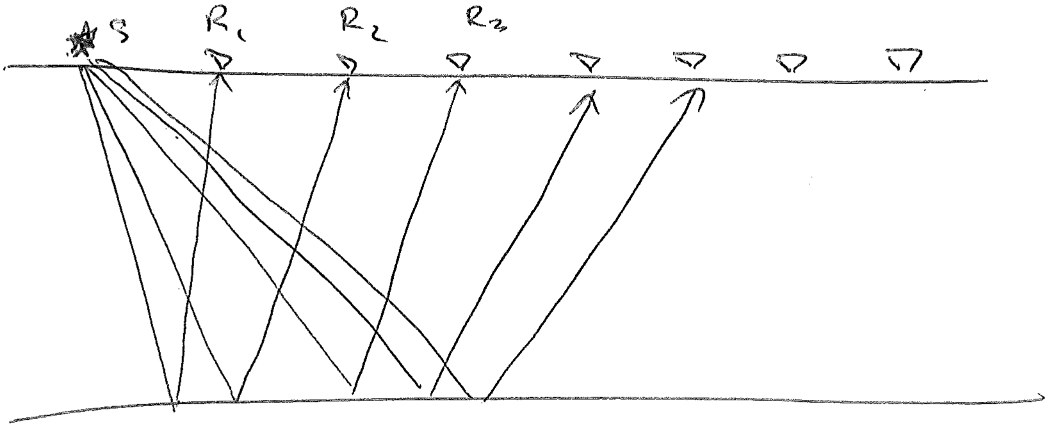


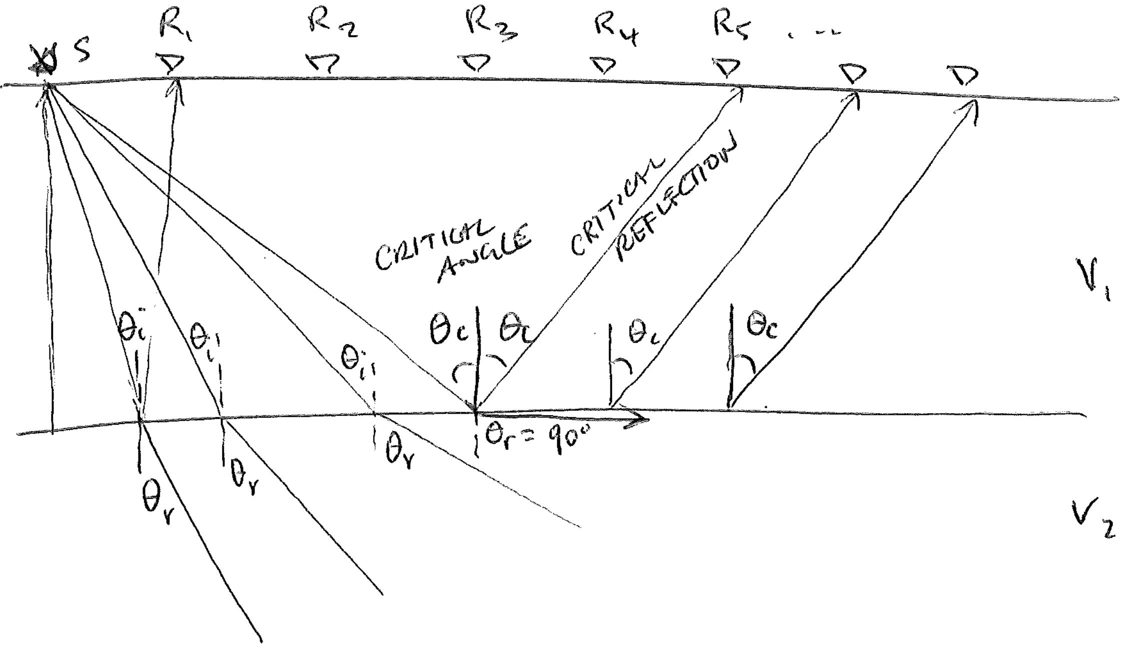
$V_{DIRECT} > V_{AIR} > V_{GROUND\ ROLL}$

$V_2 > V_1$

TRAVEL-TIME DIAGRAM







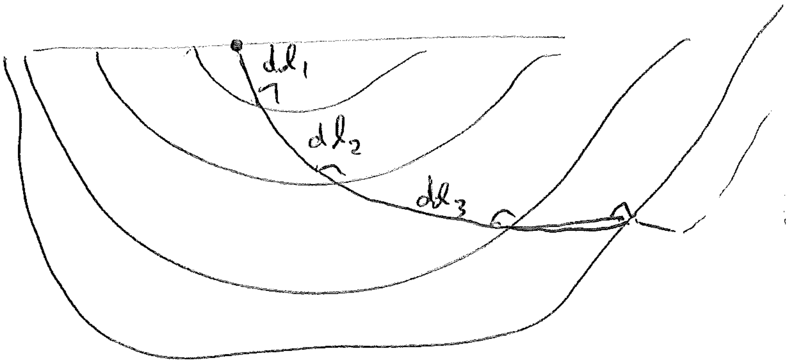
LENGTH OF PATH.

$$t = \frac{l}{v}$$

VELOCITY

travel time

$$t = \min \int_S^R \frac{dl}{V(x,y,z)}$$

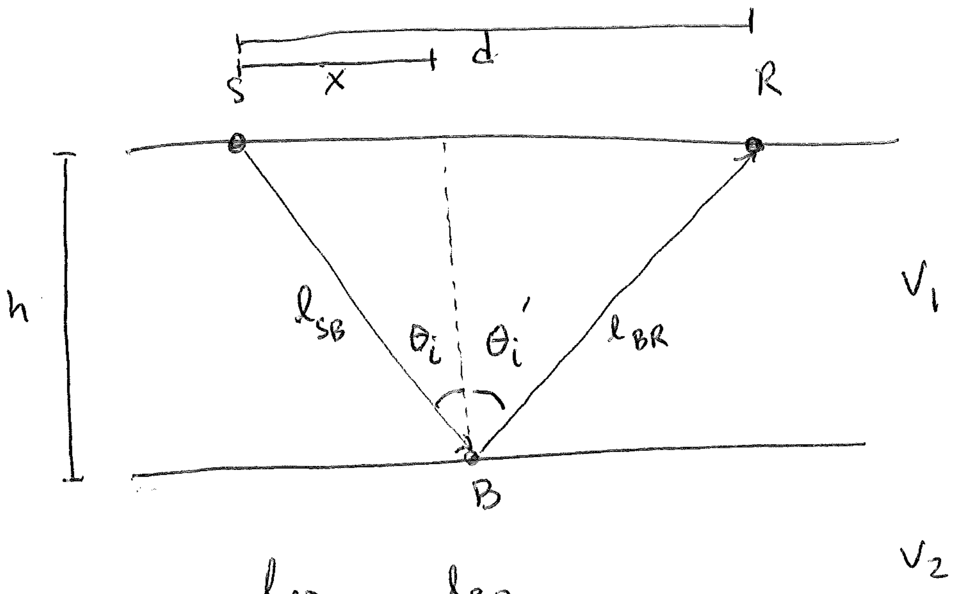


REFLECTION

ISOTROPIC

NO VELOCITY GRADIENTS

HORIZONTAL INTERFACES



$$t = \frac{l_{SB}}{V_1} + \frac{l_{BR}}{V_1}$$

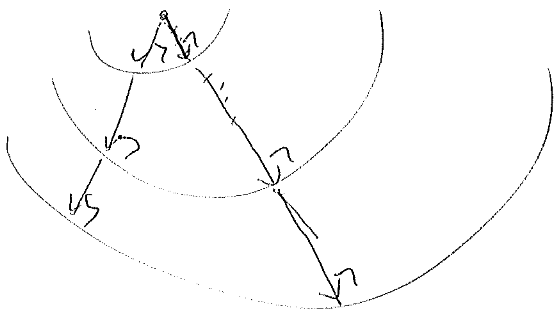
$$l_{SB} = (x^2 + h^2)^{1/2} \qquad l_{BR} = ((d-x)^2 + h^2)^{1/2}$$

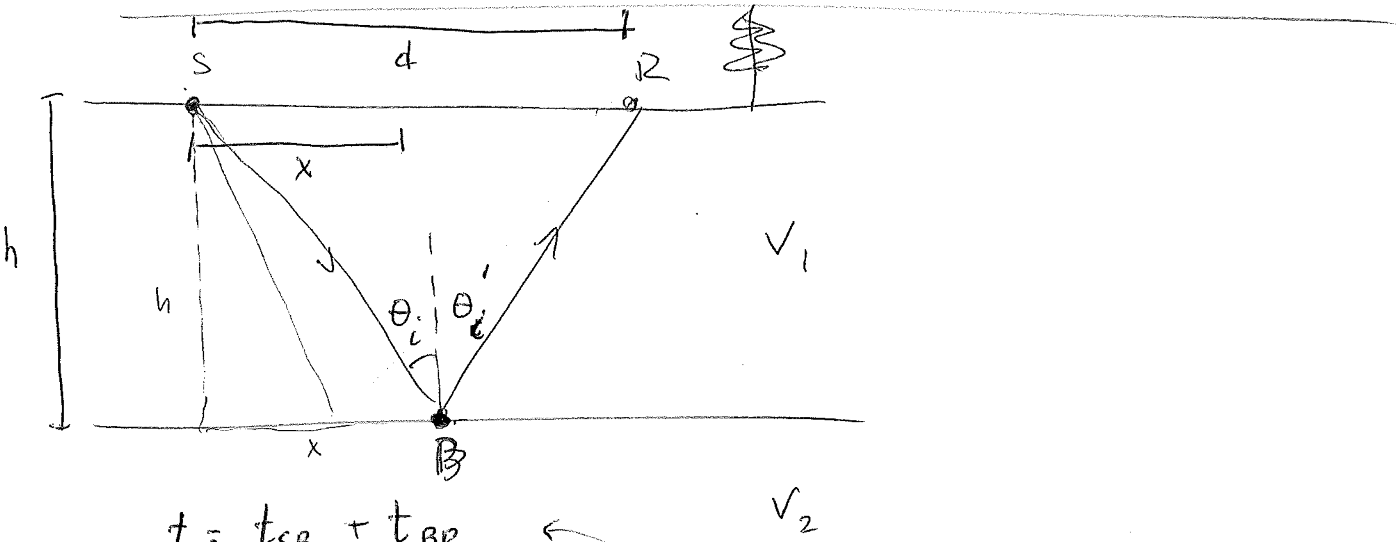
$$\min\left[t = \frac{(x^2+h^2)^{1/2}}{V_1} + \frac{((d-x)^2+h^2)^{1/2}}{V_1} \right]$$

$$\frac{dt}{dx} = 0 = \frac{1}{V_1}\left[\frac{x}{(x^2+h^2)^{1/2}} - \frac{(d-x)}{((d-x)^2+h^2)^{1/2}} \right]$$

$$t = \frac{l}{V}$$

$$t = \min \int_S^R \frac{dl}{V(x,y,z)}$$





$$t = t_{SB} + t_{BR}$$

TOTAL TRAVEL-TIME IS TIME OF EACH SEGMENT.

$$t = \frac{l_{SB}}{V_1} + \frac{l_{BR}}{V_1}$$

$$\boxed{\min t = \frac{(x^2+h^2)^{1/2}}{V_1} + \frac{((d-x)^2+h^2)^{1/2}}{V_1}}$$

TO FIND MINIMUM, TAKE DERIVATIVE, SET TO ZERO.

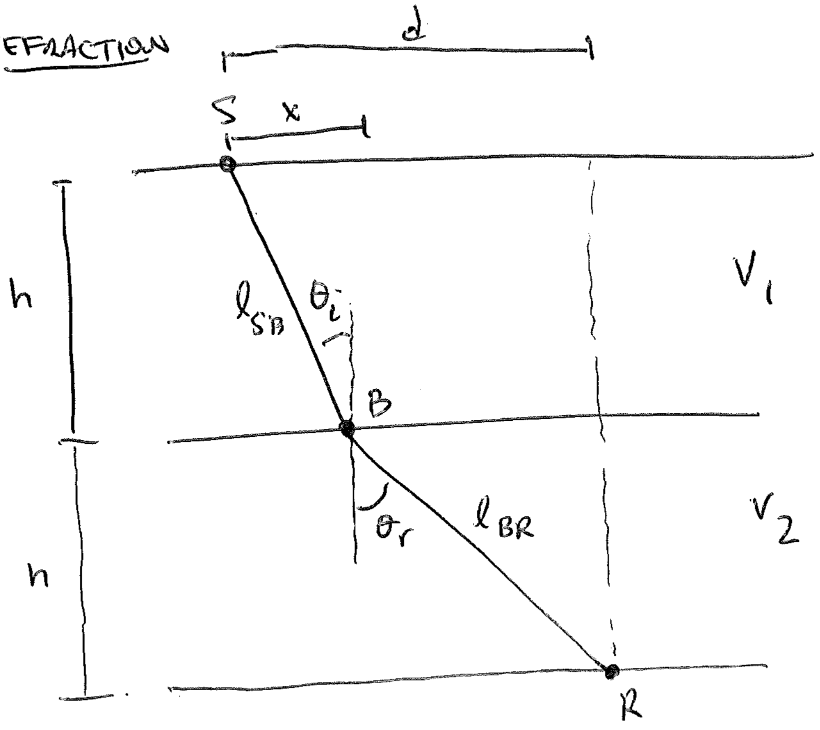
$$\frac{dt}{dx} = 0 = \frac{1}{V_1}\left[\frac{x}{(x^2+h^2)^{1/2}} - \frac{(d-x)}{((d-x)^2+h^2)^{1/2}}\right]$$

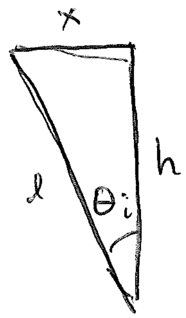
$$\frac{(d-x)}{((d-x)^2+h^2)^{1/2}} = \frac{x}{(x^2+h^2)^{1/2}}$$

$$\frac{(d-x)^2}{(d-x)^2+h^2} = \frac{x^2}{x^2+h^2}$$

$$(x^2+h^2)(d-x)^2 = x^2((d-x)^2+h^2)$$

REFRACTION





$$l^2 = h^2 + x^2$$

$$\sin\theta_i = \frac{x}{l} = \frac{x}{(h^2+x^2)^{1/2}}$$

$$t = t_{SB} + t_{BR}$$

$$t = \frac{l_{SB}}{V_1} + \frac{l_{BR}}{V_2}$$

$$l_{SB} = (x^2+h^2)^{1/2} \qquad l_{BR} = ((d-x)^2+h^2)^{1/2}$$

$$\min\left[t = \frac{(x^2+h^2)^{1/2}}{V_1} + \frac{((d-x)^2+h^2)^{1/2}}{V_2}\right]$$

$$\frac{dt}{dx} = 0 = \frac{1}{V_1}\left(\frac{x}{(h^2+x^2)^{1/2}} - \frac{1}{V_2}\frac{d-x}{(h^2+(d-x)^2)^{1/2}}\right)$$

AGAIN, TAKE DERIVATIVE SET TO ZERO

$$= \frac{1}{V_1}\sin\theta_i - \frac{1}{V_2}\sin\theta_r$$

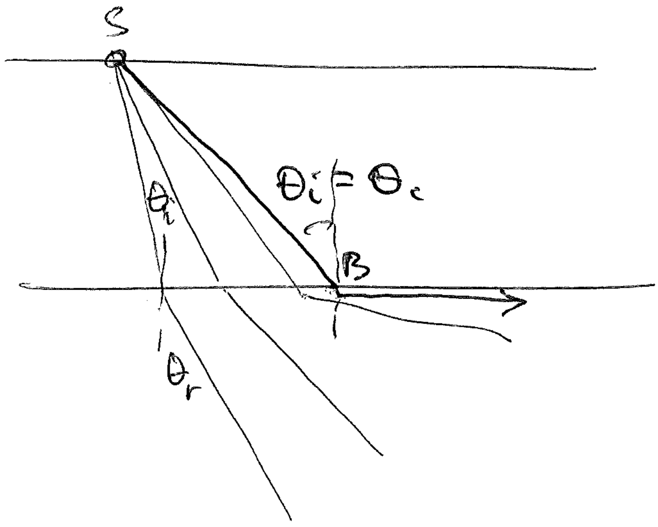
$$\frac{\sin\theta_r}{V_2} = \frac{\sin\theta_i}{V_1} \Rightarrow \text{SNELL'S LAW.}$$

CRITICAL ANGLE, $\theta_c \ \theta_r = 90°$

$$\frac{1}{V_2} = \frac{\sin\theta_c}{V_1}$$

$$\sin\theta_c = \frac{V_1}{V_2}$$

CRITICAL ANGLE, θ_c

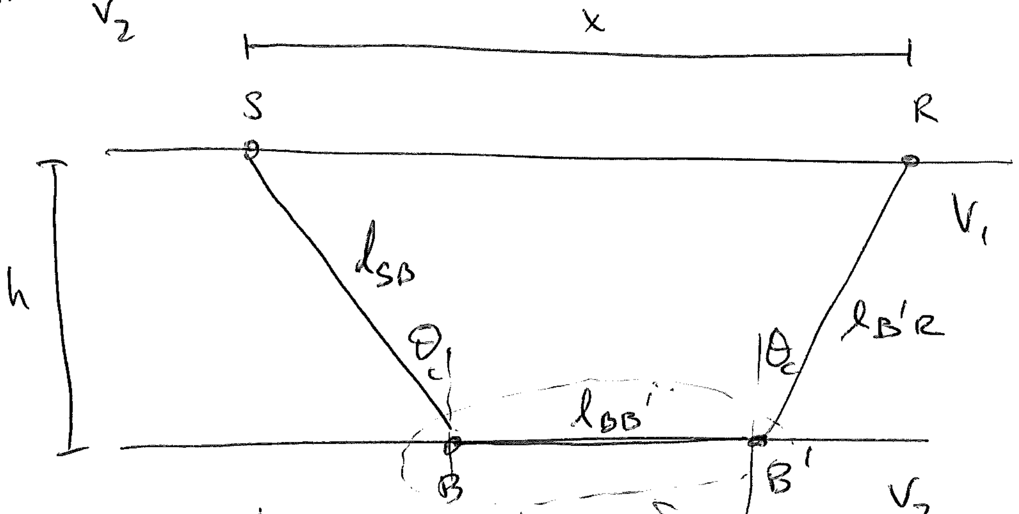


$$\theta_r = 90^\circ$$

$$\frac{1}{V_2} = \frac{\sin\theta_c}{V_1}$$

CRITICAL REFRACTION

$$\sin^{-1}\frac{V_1}{V_2} = \theta_c$$



$$t = t_{SB} + t_{BB'} + t_{B'R}$$

$$t = \frac{l_{SB}}{V_1} + \frac{l_{BB'}}{V_2} + \frac{l_{B'R}}{V_1}$$

WAVE TRAVELS JUST BELOW INTERFACE

$\Rightarrow V = V_2$

RELECTION / TRANSMISSION COEFFICIENTS

NORMAL INCIDENT WAVE

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \leftarrow \text{ACOUSTIC IMPEDANCE} \qquad Z = \rho V$$

ρ ↑ DENSITY, V ↑ VELOCITY.

↑ REFLECTION COEFFICIENT

$$R = \frac{\rho_2 V_2 - \rho_1 V_1}{\rho_2 V_2 + \rho_1 V_1}$$

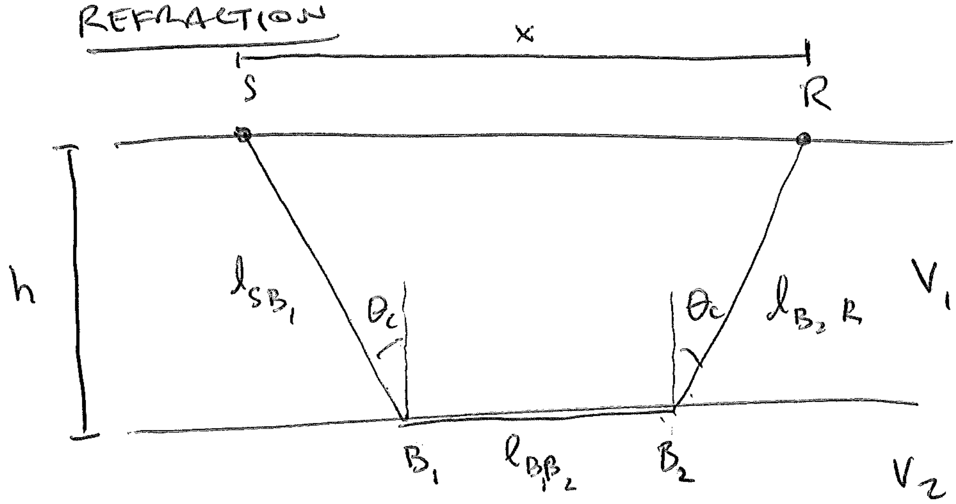
$R \to 1$ ALL ENERGY REFLECTED

$R \to 0$ ALL ENERGY IS TRANSMITTED

$$T = \frac{2Z_1}{Z_2 + Z_1}$$

↑ TRANSMISSION COEFFICIENT

REFRACTION



$$t = \frac{l_{SB_1}}{V_1} + \frac{l_{B_1 B_2}}{V_2} + \frac{l_{B_2 R}}{V_1}$$